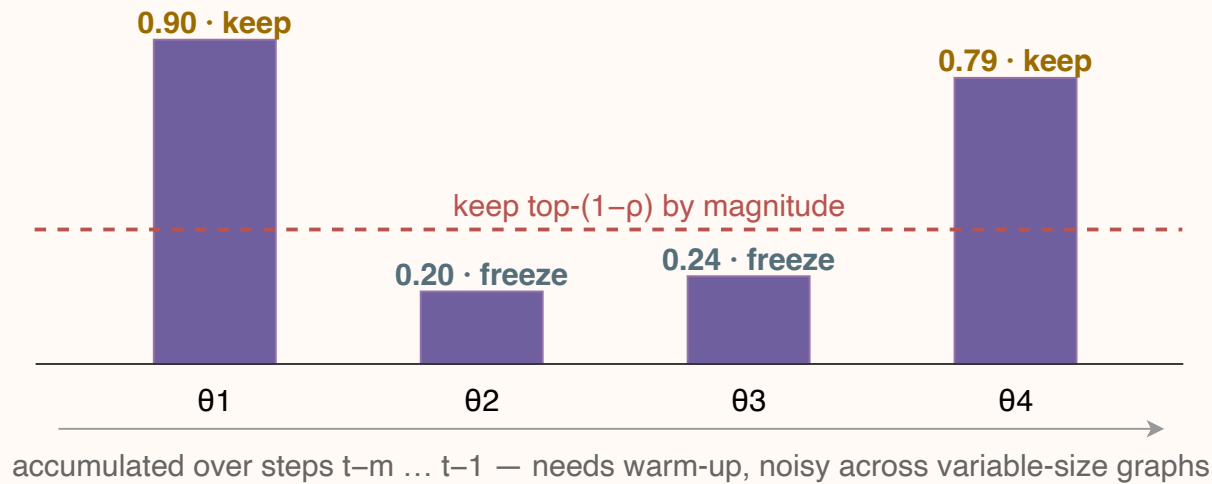


Statistical Pruning vs. QFI-Pruning in QGCN/QGAT — what the QFI metric changes

RY(x) embedding updated (SP) updated (QFI) frozen thresholds (ρ / $\kappa\lambda_{\max}$ / τ)

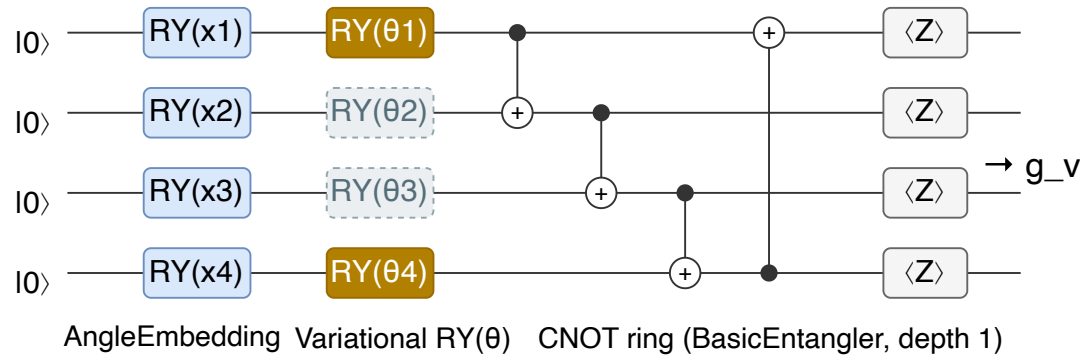
(a) Statistical Pruning (prior Q-Drop SP) — importance from gradient SLOPE, sliding window

① Importance = accumulated gradient magnitude
 $\text{score}_i = (1/m) \cdot \sum |\nabla \theta_i|$ over the last m training steps



freeze { θ_2, θ_3 }

② QGCN/QGAT node-embedding circuit (shared ϕ , applied to every node v)
 SP verdict: update { θ_1, θ_4 } • freeze { θ_2, θ_3 }

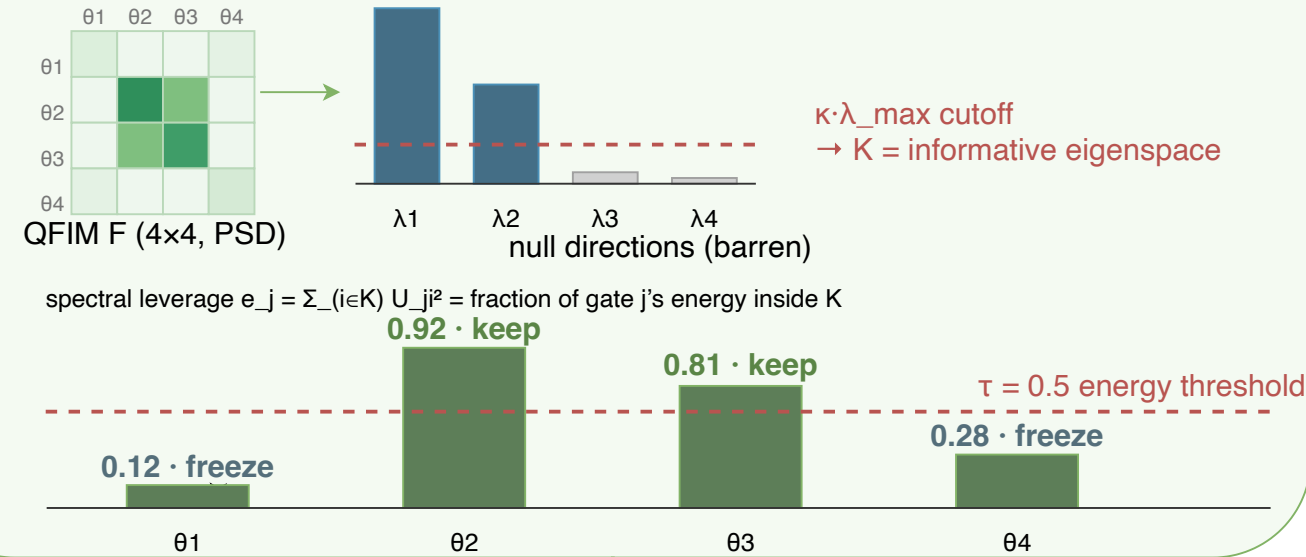


one shared circuit per node → one shared mask per mini-batch (permutation-equivariant)

③ Update — plain step on survivors: $\theta_i \leftarrow \theta_i - \eta \cdot g_i$. Raw gradient ignores curvature; kept gates can still sit in flat regions.

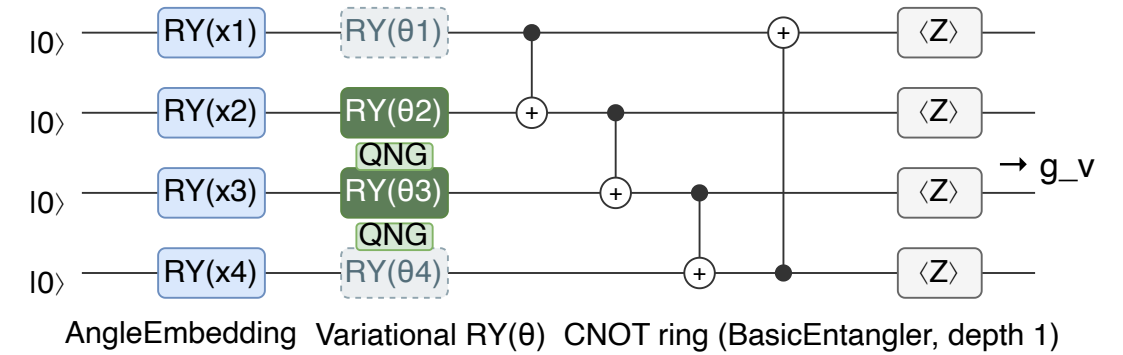
(b) QFI-Pruning (ours) — importance from information GEOMETRY, one probe per epoch

① Importance = spectral leverage on the Quantum Fisher Information Matrix
 $F = \text{mean}_v F_{\text{FS}}(z_v)$ eigen-spectrum of F



freeze { θ_1, θ_4 }

② QGCN/QGAT node-embedding circuit (same ansatz, same nodes)
 QFI verdict: update { θ_2, θ_3 } with QNG • freeze { θ_1, θ_4 }



same probe batch drives QFIM → mask shared by every node (permutation-equivariant)

③ Update — natural-gradient step on survivors: $g_S \leftarrow (F_{SS} + \epsilon I)^{-1} g_S$, $\theta_S \leftarrow \theta_S - \eta \cdot g_S$. Rescaled by the information metric of the kept sub-manifold.